AHMED IPRAHIM ABDELHAMED

Junior Data Scientist

Cairo, Egypt | +20 102 009 4823 | ahmediprahim750@gmail.com

LinkedIn: linkedin.com/in/ahmed-iprahim-a51240133 **| GitHub:** github.com/afadamarcello-code **| Portfolio:** afadamarcello-code.github.io

SUMMARY

Results-driven **Data Scientist** with a B.Sc. in Computer Science and ALX certification in **Machine Learning and Data Science**. Experienced in building end-to-end ML models including **hybrid recommender systems, regression pipelines, classification algorithms**, and predictive analytics solutions. Strong foundation in **Python, Scikit-learn, XGBoost, SQL**, and statistical modeling, with hands-on experience applying data-driven approaches to fintech operations at Tabby.

SKILLS

- **Machine Learning & Modeling:** Supervised/Unsupervised Learning, Feature Engineering, XGBoost, Regression/Classification, SVD, Hyperparameter Tuning
- **Languages & Querying:** Python (Pandas, NumPy, Scikit-Learn, Statsmodels), SQL (PostgreSQL, DuckDB), Bash, Git
- **Visualization & Deployment:** Streamlit, Power BI, Looker Studio, Plotly, Matplotlib, Seaborn
- **Data Engineering & Diagnostics:** ETL Pipelines, EDA, VIF Analysis, LINE Framework, Statistical Testing

PROJECTS

Credit Risk & Expected Loss AI Engine *| Python, XGBoost, PostgreSQL, Streamlit, Scikit-learn*

- Engineered an end-to-end classification pipeline predicting applicant **Probability of Default (PD) on 32,000+ financial records** while preventing data leakage by dropping post-outcome metrics.
- Handled severe class imbalance (78% good/22% default) using **XGBoost scale_pos_weight and GridSearchCV**, boosting default recall to **78% without sacrificing model precision**.
- Deployed the trained model as an **interactive Streamlit web application**, enabling real-time credit scoring and automated risk classification for new loan applicants.

CineMatch AI – Hybrid Movie Recommender System *| Python, Scikit-learn, Pandas, NumPy, Streamlit*

- Built an end-to-end hybrid recommender processing **10M+ MovieLens ratings** using SVD (bias + latent factors).
- Achieved an **RMSE of 0.8794**, outperforming baseline models by **17%** through feature engineering and hyperparameter tuning.
- Deployed an **interactive Streamlit web application** featuring fuzzy string search and real-time similarity matching.

Ames Housing Valuation & Regression Diagnostics *| Python, Scikit-learn, Statsmodels, Streamlit*

- Built an end-to-end regression pipeline predicting real estate valuations while optimizing model complexity and signal density.
- Reduced feature space from **37 to 24** by diagnosing correlation matrices and Variance Inflation Factors (VIF) to **eliminate 13 multicollinear variables**.
- Validated **LINE regression assumptions** and engineered 4 interaction/polynomial features, boosting **R^2 from 0.8356 to 0.8999**.
- Deployed an **interactive Streamlit application** enabling real-time valuation estimates for stakeholders.

PROFESSIONAL EXPERIENCE

Tabby | Data & Analytics Specialist (01/2024 – Present)

- Built Python scripts to extract, clean, and aggregate multi-channel social engagement datasets of over **50,000+ records** for quantitative performance analysis.
- Automated weekly reporting pipelines using Python and SQL, eliminating manual data collection and **reducing reporting time by 10+ hours per week**.
- Performed time-series and statistical trend analysis on **10,000+ weekly customer inquiry volumes**, accurately forecasting demand spikes and optimizing operational capacity.

Tabby | Social Media & Customer Support Specialist (01/2023 – 12/2023)

- Analyzed historical performance metrics across chat, email, and phone channels to identify service level variance, consistently maintaining **92%+ CSAT benchmark**.
- Managed real-time interaction datasets and pattern anomalies to maintain SLA adherence during volume spikes.

EDUCATION & CERTIFICATIONS

Helwan University | Bachelor of Science in Computer Science | 2024 | Helwan, Egypt

Certifications: Data Science Certificate (ALX-2024) | Machine Learning Certificate (ALX-2024) | Data Analytics Professional Certificate (ALX-2024) | Python Programming Certificate (ALX-2024)